An Algorithm for Predicting the Intelligibility of Speech Masked by Modulated Noise Maskers

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***Abstract*—Intelligibility listening tests are necessary during development and evaluation of speech processing algorithms, despite the fact that they are expensive and time consuming. In this paper, we propose a monaural intelligibility prediction algorithm, which has the potential of replacing some of these listening tests. The proposed algorithm shows similarities to the short-time objective intelligibility (*STOI*) algorithm, but works for a larger range of input signals. In contrast to STOI, extended STOI (*ESTOI*) does not assume mutual independence between frequency bands. ESTOI also incorporates spectral correlation by comparing complete 400-ms length spectrograms of the noisy/processed speech and the clean speech signals. As a consequence, *ESTOI* is also able to accurately predict the intelligibility of speech contaminated by temporally highly modulated noise sources in addition to noisy signals processed with time–frequency weighting. We show that ESTOI can be interpreted in terms of an orthogonal decomposition of short-time spectrograms into intelligibility subspaces, i.e., a ranking of spectrogram features according to their importance to intelligibility. A free MATLAB implementation of the algorithm is available for noncommercial use at http://kom.aau.dk/∼jje/.**

***Index Terms*—Modulated noise sources, noise reduction, objective distortion measures, speech enhancement, speech intelligibility prediction.**

I. Introduction

When developing speech communication systems for human receivers, listening tests play a major role both for monitoring progress in the development phase and for verifying the performance of the final system. Often, listening tests are used to quantify aspects of speech quality and speech intelligibility. Although listening tests constitute the only tool available for measuring ground-truth end-user impact, they are time-consuming, they may require special auditory stimuli data and test equipment, and they require the availability of a group of typical end-users. For these reasons, listening tests are costly and can typically not be employed many times during the development phase of a speech communication system. Hence, cheaper alternatives or supplements are of interest.

In this paper, we focus on intrusive, monaural intelligibility prediction models, i.e., algorithms which—rather than conducting an actual listening test—predict the outcome of the listening test based on the auditory stimuli of the test. Historically, two lines of research serve as the foundation for existing intelligibility prediction models: i) the *Articulation Index (AI)* [1] by French and Steinberg [2], which was later refined and standardized as the *Speech Intelligibility Index (SII)* [3], and ii) the *Speech Transmission Index (STI)* [4] by Steeneken and Houtgast [5].

AI and *SII* were developed with simple linear signal degradations, e.g., additive noise, in mind. To estimate intelligibility, the methods divide the signal under analysis into frequency subbands and assume that each subband contributes independently to intelligibility. The contribution of a subband is found by estimating the long-term speech and noise power within the subband to arrive at the long-term subband signal-to-noise ratio (SNR). Then, subband SNRs are limited to the range from -15 to $+15$ dB, normalized to a value between 0 and 1, and combined as a perceptually weighted average.

STI extends the range of distortions to include convolutive noise, e.g., reverberant speech and effects of room acoustics. *STI* is based on the observation that reverberation and/or additive noise tend to reduce the depth of temporal signal modulations compared to the clean, undistorted reference signal. To measure changes in the modulation transfer function, *STI* generates bandpass filtered noise probe signals at different center-frequencies, and amplitude-modulates each such signal at different modulation frequencies relevant to speech intelligibility. Each modulated probe signal is then passed through the communication channel in question (e.g., characterised by a room impulse response), and the reduction in modulation depth is finally translated into an intelligibility index.

Despite the importance of *AI* and *SII*, the methods have a number of limitations. First, they require the long-term spectrum of the additive noise signal to be known in advance. Secondly, since the methods rely on long-term statistics, they cannot discern modulated noise signals from un-modulated ones, when their long-term spectra are identical. In other words, the intelligibility of speech contaminated by modulated and un-modulated noise is judged to be identical, although it is well-known that this is generally not the case, e.g., [6], [7]. Finally, *AI* and *SII* are not directly applicable to signals which have been passed through some non-linear processing stage before presentation to the listener, because in this case it is no longer clear which noise spectrum to use.

Various methods have been proposed to reduce the limitations mentioned above and extend the range of acoustic situations for which intelligibility prediction can be made. In [6], Rhebergen and Versfeld proposed the *Extended SII (ESII)*, which avoids

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the use of long-term noise spectra in *SII*. Specifically, *ESII* divides the masker signal into short time frames (9–20 ms) and averages the SII computed for each frame individually, to predict intelligibility for fluctuating noise sources. In a somewhat similar manner, the *Glimpse Model* by Cooke [8] uses realizations of speech and additive noise signals to estimate the glimpse percentage, i.e., the fraction of time-frequency tiles whose SNR exceeds a certain threshold, which is then translated to an intelligibility estimate. The *Coherence SII (CSII)* by Kates and Arehart [9] extends *SII* to better take into account various non-linear distortions, including center- and peak-clipping. Similarly, Goldsworthy and Greenberg [10] proposed a modified STI approach, *speech STI*, which replaces the traditional noise probe signals with actual speech signals. This was done to better take into account the effect of non-linear distortions such as envelope clipping [11], and dynamic amplitude compression [12].

More recently, methods have emerged which are inspired by both original lines of research. For example, Jørgensen and Dau [13] decompose the speech signal and an additive noise masker in a modulation filter bank. Intelligibility prediction is then based on the envelope power SNR at the output of this filter bank. Taal *et al.* [14] proposed the *Short-Time Objective Intelligibility (STOI)* measure, which extracts temporal envelopes of undistorted and noisy/processed speech signals in frequency subbands. The envelopes are then subject to a clipping procedure, compared using short-term linear correlation coefficients, and a final intelligibility prediction is constructed simply as an average of the correlation coefficients. *STOI* has proven to be able to predict quite accurately the intelligibility of speech in many acoustic situations, including the speech output of mobile phones [15], noisy speech processed by ideal time-frequency masking and single-channel speech enhancement algorithms [14], speech processed by cochlear implants [16], and *STOI* appears robust to different language types including Danish [14], Dutch [17], and Mandarin [18]. Although *STOI* performs well in many cases, some of the algorithmic choices, e.g., the use of a linear correlation coefficient as a basic distance measure, are less well motivated from a theoretical point of view. However, in [17] Jensen and Taal proposed the *Speech Intelligibility prediction based on Mutual Information (SIMI)* method, which suggests that characteristics of *STOI* may be explained using information theoretic arguments.

STOI, and several of the methods described above, show only weak links to the properties of the auditory system, and much more elaborate models have been proposed, e.g., [19], [20]. For example, the Hearing-Aid Speech Perception Index by Kates and Arehart [20] employs level- and hearing-profile dependent auditory filterbanks to compute quantities resembling mel-frequency cepstral coefficients (MFCCs) for the clean reference signal and the noisy/processed signal, respectively. Then, long-term correlations between clean and noisy/processed MFCCs are computed, before the average across the cepstral dimension is found. Finally, this cepstral correlation average is combined with estimates of the auditory signal coherence for low-, mid-, and high-intensity signal regions [9], to form an intelligibility index.

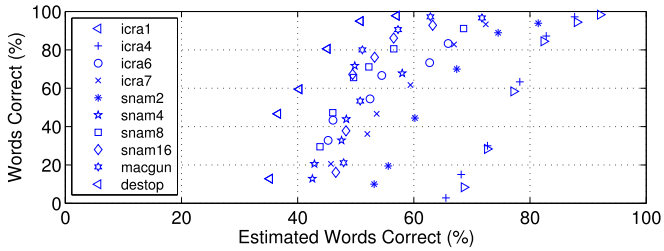


Fig. 1. Measured versus predicted intelligibility (*STOI* [14]) for speech in ten different additive, modulated noise sources (6 SNRs each). Linear correlation coefficient between measured and predicted intelligibility is $\rho = 0.47$. For more information on noise sources, see Table I.

Existing intelligibility prediction methods may be divided into two classes: 1) methods which require that the target speech signal and the distorting component (e.g., the additive noise) are available in separation, e.g., [3], [4], [6], [8], [13] and 2) methods which do not impose this requirement, e.g., [9], [10], [14], [17]. Class-1 methods have the advantage that they can use the access to speech and noise realizations to compute SNR realizations in different time-frequency regions and find an intelligibility index based on these. Class-2 methods, on the other hand, cannot observe SNRs directly, but must rely on features estimated from, generally, limited data, e.g., short-time correlations estimated from the noisy/processed and clean speech signal. The disadvantage of Class-1 methods is that they are not applicable to non-linearly processed noisy speech signals, because in this situation noise and speech signals are not readily available in separation. Class-2 methods are more generally applicable, i.e., also to noisy signals, which have been non-linearly processed.

As reported in [21], *STOI*—and as we show in Section IV—other Class-2 methods have limitations for target speech signals in additive noise sources with strong temporal modulations, as e.g. a single competing speaker. To demonstrate this point, Fig. 1 shows an example of intelligibility predicted by *STOI* vs. actual intelligibility, measured in listening tests with speech signals degraded by 10 highly modulated masker signals at different SNRs (details are given in Table I and will be discussed later). Clearly, *STOI* performs less well in this situation: the linear correlation coefficient between predicted and measured intelligibility is as low as $\rho = 0.47$.

In this paper, we propose a new Class-2 intelligibility predictor—Extended Short-Time Objective Intelligibility (ESTOI)—which, unlike many existing Class-2 methods, works well for highly modulated noise sources as the example above.[1] Importantly, *ESTOI* also works well in situations, where existing Class-2 methods work well.

As the name suggests, *ESTOI* is inspired by *STOI* [14]. As with *STOI*, *ESTOI* operates within a 384 ms analysis window on amplitude envelopes of subband signals. This analysis window is used in order to include important temporal modulation frequencies relevant for speech intelligibility [14]. To understand the differences between *STOI* and *ESTOI*, let us first

[1]The algorithm name *ESTOI* follows the terminology introduced by Rhebergen and Versfeld, who used the name ESII for their algorithm to improve the performance of *SII* for modulated noise maskers [6].

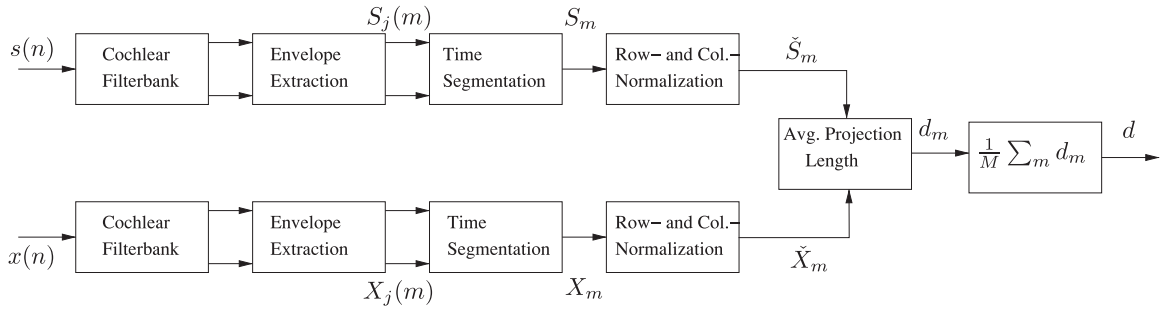


Fig. 2. The proposed intelligibility predictor, *ESTOI*, is a function of the noisy/processed signal $x(n)$ and clean speech signal $s(n)$. First, the signals are passed through a one-third octave filter bank, and the temporal envelopes of each subband signal are extracted. The resulting clean and noisy/processed short-time envelope spectrograms are time- and frequency-normalized before the "distance" between them is computed, resulting in intermediate, short-time intelligibility indices d_m. Finally, the intermediate indices are averaged to form the final intelligibility index d. More details are given in Section II. Signal examples of the various stages are shown in Fig. 3.

interpret, how *STOI* computes a correlation coefficient for a 384 ms analysis window. *STOI* first computes linear correlation coefficients for each subband between undistorted and noisy/processed signals[2]; this is equivalent to computing inner products between mean- and variance- normalized envelope signals. To find a correlation coefficient for the 384 ms analysis window, *STOI* then averages these temporal correlation coefficients across frequency, an operation which implies independent frequency band contributions to intelligibility, and which is not in line with literature, e.g., [22]. *ESTOI* shares the first step with *STOI*: mean- and variance-normalization is applied to subband envelopes. However, rather than computing the average inner products between these normalized envelopes, and relying on the additive-intelligibility-across-frequency assumption as is done in *STOI*, *ESTOI* instead computes *spectral* correlation coefficients, which are finally averaged across time within the 384 ms analysis segment. This allows *ESTOI* to better capture the effect of time-modulated noise maskers, where spectral correlation 'in the dips' is often preserved [23]. We show that *ESTOI* may be interpreted in terms of an orthogonal decomposition of energy-normalized spectrograms, i.e., a decomposition into "intelligibility subspaces" which are each ranked according to their (estimated) contribution to intelligiblity. This decomposition is important in understanding *ESTOI* and in linking its performance to perceptual studies with human listeners. Specifically, analysis of the spectrograms related to each intelligibility subspace shows that the spectro-temporal modulation frequencies, which are judged by *ESTOI* to be important for speech intelligibility, agree with the results of experimental studies of human sensitivity to spectro-temporal modulations [24].

The paper is structured as follows. In Section II we describe the *ESTOI* predictor. In Section III we interpret the predictor in terms of orthogonal intelligibility subspaces. Section IV evaluates the performance of *ESTOI*, and compares it to a range of existing algorithms. Finally, Section V concludes the work.

[2]We ignore the clipping procedure used in *STOI* in this description.

II. Proposed Model

The overall structure of the proposed intelligibility predictor, *ESTOI*, is outlined in Fig. 2. *ESTOI* is a function of the noisy/processed signal under study $x(n)$, and the clean, undistorted speech signal $s(n)$. The goal of *ESTOI* is to produce a scalar output d, which is monotonically related to the intelligibility of $x(n)$.

A. Time-Frequency Normalized Spectrograms

Let us assume that $s(n)$ and $x(n)$ are perfectly time-aligned, and that regions where $s(n)$ shows no speech activity (e.g., pauses between sentences) have been removed from both signals. In the following, we present expressions related to the clean signal $s(n)$; similar expressions hold for the noisy/processed signal $x(n)$. Let $S(k,m)$ denote the short-time Fourier transform (STFT) of $s(n)$, that is

$$\bar{s}_{j,m} = \frac{1}{\|(s_{j,m} - \mu_{s_{j,m}}\underline{1})\|}\left(s_{j,m} - \mu_{s_{j,m}}\underline{1}\right),$$

where k and m denote the frequency bin index and the frame index, respectively, and D and N' denote the frame shift in samples and FFT order, respectively. Finally, $w(n)$ is an analysis window.

To model crudely the signal transduction in the cochlear inner hair cells, a one-third octave band analysis is approximated by summing STFT coefficient energies,

$$S_j(m) = \sqrt{\sum_{k \in CB_j} |S(k,m)|^2}, \qquad j = 1, \ldots, J,$$

where j is the one-third octave band index, CB_j denotes the index set of STFT coefficients related to the jth one-third octave frequency band, and J denotes the number of subbands.

Let us collect spectral values $S_j(m)$ for each frequency band $j = 1, \ldots, J$, and across a *time segment* of N spectral samples, and arrange these in a short-time spectrogram matrix

$$S_m = \begin{bmatrix} S_1(m-N+1) & \cdots & S_1(m) \\ \vdots & & \vdots \\ S_J(m-N+1) & \cdots & S_J(m) \end{bmatrix}.$$

Hence, the jth row of S_m represents the temporal envelope of the signal in subband j. Typical parameter choices are $J = 15$ and $N = 30$ (corresponding to 384 ms) [14]. The noisy/processed short-time spectrogram matrix X_m is defined analogously.

ESTOI operates on mean- and variance- normalized rows and columns of S_m (and X_m) as follows. Let

$$s_{j,m} = [S_j(m - N + 1)\, S_j(m - N + 2) \cdots S_j(m)]^T$$

denote the jth row of the spectrogram matrix S_m. The jth mean- and variance- normalized row of S_m is given by

$$\bar{s}_{j,m} = \frac{1}{\|(s_{j,m} - \mu_{s_{j,m}} \underline{1})\|} \left(s_{j,m} - \mu_{s_{j,m}} \underline{1}\right), \tag{1}$$

where $\|y\| = \sqrt{y^T y}$ is the vector 2-norm, $\underline{1}$ is an all-one vector, and $\mu_{s_{j,m}}$ is the sample mean given by

$$\mu_{s_{j,m}} = \frac{1}{N} \sum_{m'=0}^{N-1} S_j(m - m'). \tag{2}$$

Note that the sample mean and variance of the elements in vector $\bar{s}_{j,m}$ is zero and one, respectively. The mean and variance-normalized rows $\bar{x}_{j,m}$ of the noisy/processed signal are defined similarly.

As mentioned, this row-normalization procedure is similar to the one used in *STOI*. Specifically, *STOI* uses an intermediate temporal correlation coefficient for the jth subband in the mth time segment, which can be expressed as the inner product of normalized vectors,

$$\bar{s}_{j,m}^T \bar{x}_{j,m}. \tag{3}$$

However, as mentioned, in *ESTOI* we do not use Eq. (3) directly, but introduce a spectral normalization as follows. Let us first define the row-normalized spectrogram matrix

$$\bar{S}_m = \begin{bmatrix} \bar{s}_{1,m}^T \\ \vdots \\ \bar{s}_{J,m}^T \end{bmatrix}.$$

Then, let $\check{s}_{n,m}$ denote the mean- and variance- normalized nth column, $n = 1, \ldots, N$ of matrix $\bar{S}_m$, where the normalization is carried out analogously to Eqs. (1) and (2). We finally define the row- and column-normalized matrix $\check{S}_m$ as

$$\check{S}_m = [\check{s}_{1,m} \;\cdots\; \check{s}_{N,m}].$$

Hence, the columns of $\check{S}_m$ represent unit-norm, zero-mean normalized spectra (which themselves are computed from normalized temporal envelopes). The row- and column-normalized matrix $\check{X}_m$ of the noisy/processed signal in time segment m is defined in a similar manner. Fig. 3 demonstrates the effect of the various normalizations on example clean and noisy spectrograms.

B. Intelligibility index

The row- and column- normalized matrices $\check{S}_m$ and $\check{X}_m$ serve as the basis for the proposed intelligibility predictor. In particular, we define an *intermediate* intelligibility index, related

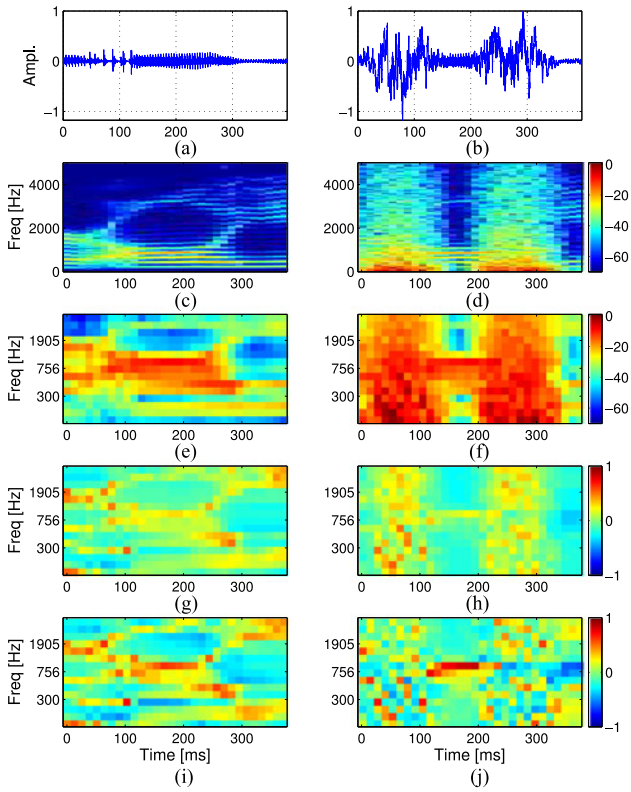


Fig. 3. Short-time spectrograms for clean speech time segment (left column) and noisy time segment (right column) for additive, speech-shaped, sinusoidal amplitude-modulated Gaussian noise (modulation frequency of 5 Hz, SNR $= -10$ dB). (a), (b) Time domain segments. (c), (d) DFT short-time spectrograms $|S(k,m)|, |X(k,m)|$ (dB scale), computed by applying an $N' = 512$ point FFT to zeropadded, Hann windowed, time-domain frames of 256 samples (25.6 ms) with an overlap of $D = 128$ samples. (e), (f) Third-order octave filterbank spectrograms S_m, X_m (dB scale). (g), (h) Spectrograms with mean- and variance-normalized rows $\bar{S}_m$, $\bar{X}_m$ (linear scale). (i), (j) Spectrograms with mean- and variance-normalized rows and columns, $\check{S}_m$, $\check{X}_m$ (linear scale).

to time segment m, simply as

$$d_m = \frac{1}{N} \sum_{n=1}^{N} \check{s}_{n,m}^T \check{x}_{n,m}. \tag{4}$$

Since $\check{s}_{n,m}$ and $\check{x}_{n,m}$, $n = 1, \ldots, N$ are unit-norm vectors, each term in the sum may be recognized as the (signed) length of the orthogonal projection of the noisy/processed vector $\check{x}_{n,m}$ onto the clean vector $\check{s}_{n,m}$ or vice versa. It follows that $-1 \leq \check{s}_{n,m}^T \check{x}_{n,m} \leq 1$. Similarly, d_m may be interpreted as the (signed) length of these projections, averaged across time within a time segment. In low-noise situations where $\check{x}_{n,m} \approx \check{s}_{n,m}$, then d_m will be close to its maximum average projection length of 1, whereas if the elements of $\check{x}_{n,m}$ and $\check{s}_{n,m}$ are uncorrelated, then $d_m \approx 0$, i.e., the vectors are approximately orthogonal. Also, from the definitions of $\check{s}_{n,m}$ and $\check{x}_{n,m}$, d_m may be interpreted simply as sample correlation coefficients of the columns of $\bar{S}_m$ and $\bar{X}_m$ (i.e., spectra which have been normalized according to their subband envelopes), averaged across the N frames within a segment.

For simplicity, the intelligibility index related to the entire noisy/processed signal of interest is then defined as the temporal

average of the intermediate intelligibility indices,

$$d = \frac{1}{M}\sum_{m=1}^{M} d_m, \tag{5}$$

where M is the number of time segments in the signal of interest. Since $-1 \leq d_m \leq 1$, it follows that $-1 \leq d \leq 1$.

C. Implementation

ESTOI operates at a sampling frequency of 10 kHz to ensure that the frequency region relevant for speech intelligibility is covered [2]; all signals are resampled to this frequency before applying the method. Then, signals are divided into frames of 256 samples, using a frame shift of $D = 128$, the frames are windowed with a Hann window, and an FFT of order $N' = 512$ is applied. Before computing the intelligibility index, frames with no speech content are discarded. These are identified as the frames of the reference speech signal $s(n)$ with energy less than 40 dB than the signal frame with maximum energy. DFT coefficients of speech active frames are grouped into $J = 15$ one-third octave bands, with center frequencies of 150 Hz and approximately 4.3 kHz, for the lowest and highest band, respectively. Finally, time segments of length $N = 30$ (corresponding to 384 ms) are used (for further details on this choice, we refer to Section IV-C).

III. Orthogonal Intelligibility Subspaces

In this section we present interpretations of Eqs. (4) and (5) which provide insights into *ESTOI*. Specifically, we show that *ESTOI* can be interpreted in terms of a decomposition of (row- and column- normalized) noisy/processed short-time spectrograms into orthogonal one-dimensional subspaces. The decomposition assigns an intelligibility score to each such subspace, so that the sum of all the subspace intelligibilities equals the total intermediate intelligibility d_m of the noisy/processed short-time spectrogram. The decomposition therefore allows us to rank each subspace according to their (predicted) contribution to intelligibility, revealing which spectro-temporal features are predicted to be important to intelligibility.

A. Preliminaries

To focus our exposition, we re-write the expression for d_m (see Eq. (4)) using the columns of the row- and column-normalized matrices $\check{X}_m$ and $\check{S}_m$. Let us concatenate the N columns of $\check{S}_m$ into a supervector

$$\check{s}_m = \left[\check{s}_{1,m}^T \ \ldots \ \check{s}_{N,m}^T\right]^T,$$

where $\check{s}_m \in \Re^{NJ\times 1}$. A similar definition holds for the noisy/processed supervector $\check{x}_m$. Furthermore, let us collect supervectors for each segment m as columns in super matrices. The clean speech super matrix $\check{S} \in \Re^{NJ\times M}$ is given by

$$\check{S} = [\check{s}_1, \ \ldots, \ \check{s}_M].$$

The noisy/processed matrix $\check{X}$ is defined similarly.

The intermediate intelligibility index d_m (see Eq. (4)) may then be written as

$$d_m = \frac{1}{N}\check{s}_m^T \check{x}_m, \tag{6}$$

and inserting this into Eq. (5) leads to

$$d = \frac{1}{MN}\,\mathrm{Tr}\left(\check{S}^T\check{X}\right), \tag{7}$$

where $\mathrm{Tr}(\cdot)$ denotes the matrix trace operator.

Let us introduce the following orthogonal decomposition of the noisy/processed supervectors $\check{x}_m$,

$$\begin{aligned}\check{x}_m &= \sum_{l=1}^{NJ} e_l e_l^T \check{x}_m \\ &= \sum_{l=1}^{NJ} P_l \check{x}_m,\end{aligned} \tag{8}$$

where $e_i^T e_j = \delta(i,j)$ are orthonormal vectors, and $P_l = e_l e_l^T$ is an orthogonal projection matrix onto the one-dimensional subspace spanned by e_l. Since each noisy/processed supervector $\check{x}_m$ describes a time-frequency region of $N \times J$ one-third octave spectral values, Eq. (8) provides—when the basis vectors e_l are specified—a decomposition of a noisy/processed time-frequency region into mutually orthogonal one-dimensional subspaces.

B. Intelligibility Subspace Decomposition

Our goal is to determine the orthogonal basis vectors e_l in Eq. (8), ordered according to their (estimated) impact on intelligibility: first, we find the basis vector e_1, which carries most intelligibility on average across noisy/processed spectrograms. Next, we find the basis vector e_2, orthogonal to e_1, which carries most intelligibility. This procedure is repeated for the remaining dimensions, leading to an orthogonal subspace decomposition in terms of intelligibility.

To do this, insert Eq. (8) into Eq. (7),

$$\begin{aligned} d &= \frac{1}{NM}\,\mathrm{Tr}\left(\check{S}^T\check{X}\right) \\ &= \frac{1}{2NM}\,\mathrm{Tr}\left(\check{S}^T\check{X} + \check{X}^T\check{S}\right) \\ &= \frac{1}{2NM}\,\mathrm{Tr}\left(\check{S}^T\sum_l P_l\check{X} + \check{X}^T\left(\sum_l P_l\right)^T\check{S}\right) \\ &= \frac{1}{2NM}\,\mathrm{Tr}\left(\left(\check{X}\check{S}^T + \check{S}\check{X}^T\right)\sum_l P_l\right) \\ &= \frac{1}{2NM}\sum_{l=1}^{NJ} e_l^T\left(\check{X}\check{S}^T + \check{S}\check{X}^T\right)e_l, \end{aligned} \tag{9}$$

where we used that $\mathrm{Tr}\,A = \mathrm{Tr}\,A^T$, that summation and trace are linear operators whose order can be interchanged, that $\sum_l P_l = (\sum_l P_l)^T$ is a symmetric matrix by definition, and that $\mathrm{Tr}\,ABC = \mathrm{Tr}\,CAB = \mathrm{Tr}\,BCA$.

We can now perform the orthogonal intelligibility subspace decomposition described above by solving the following sequence of problems, Step 1:

$$\max_{e_1} \frac{1}{2NM} \quad e_1^T(\check{X}\check{S}^T + \check{S}\check{X}^T)e_1, \text{ such that } e_1^T e_1 = 1.$$

Steps ($l = 2, \ldots, NJ$):

$$\max_{e_l} \quad \frac{1}{2NM} e_l^T(\check{X}\check{S}^T + \check{S}\check{X}^T)e_l \text{ such that } e_l^T e_l = 1, \text{ and } e_l \perp e_1, \ldots, e_{l-1}.$$

It may be recognized that the solution vectors e_l are the eigenvectors of the symmetric matrix $\frac{1}{2NM}(\check{X}\check{S}^T + \check{S}\check{X}^T)$. The symmetry of this matrix ensures that a) the eigenvectors are mutually orthogonal, and b) the eigenvalues are real-valued, which allows a simple ranking of subspaces according to their contribution to intelligibility.

Note that inserting Eq. (8) with the found vectors e_l in Eq. (6) allows us to express the intermediate intelligibility index d_m in terms of a sum of orthogonal intelligibility subspaces. Note also that since the lth eigenvalue λ_l of the matrix $\frac{1}{2NM}(\check{X}\check{S}^T + \check{S}\check{X}^T)$ satisfies

$$\frac{1}{2NM}(\check{X}\check{S}^T + \check{S}\check{X}^T)e_l = \lambda_l e_l,$$

then it follows that

$$\lambda_l = \frac{1}{2NM} e_l^T(\check{X}\check{S}^T + \check{S}\check{X}^T)e_l.$$

Comparison to Eq. (9) shows that

$$d = \sum_{l=1}^{NJ} \lambda_l.$$

In other words, the total estimated intelligibility of the signal in question is completely determined by the eigenvalues of the sample cross-correlation matrix $\frac{1}{2NM}(\check{X}\check{S}^T + \check{S}\check{X}^T)$. Specifically, the contribution to intelligibility by the lth subspace is given by the corresponding eigenvalue λ_l, and the sum of all eigenvalues equals the total intelligibility index.

C. Intelligibility Subspaces—Example

To demonstrate the intelligibility subspace decomposition we construct noisy (but in this example unprocessed) speech signals by adding noise to clean speech signals. Specifically, we study the impact of adding a 100% intensity-modulated, lowpass filtered noise sequence to 1680 signals from the TIMIT [25] data base. The noise is a Gaussian white noise sequence (sampling rate of $f_s = 16\,000$ Hz), filtered through a first-order IIR lowpass filter with a 3 dB cut-off frequency at approximately 80 Hz (pole location at $p = 0.97$). Then, this lowpass filtered noise is amplitude modulated by the sequence

$$a(n) = 1 + \sin(2\pi f_{\mathrm{mod}}/f_s n + \phi), \quad n = 0, \ldots, N_s,$$

with a modulation frequency of $f_{\mathrm{mod}} = 5$ Hz, N_s is the sequence length corresponding to the duration of the speech signal in question, and ϕ is a uniformly distributed random phase value, drawn independently for each sentence. The noise is scaled to form an SNR of -10 dB for each sentence.

Based on this set of noisy signals, we apply the intelligibility subspace decomposition described above. Fig. 4 (lower-right) shows the $NJ = 450$ eigenvalues λ_l in descending order for the decomposition of d. In this example, the 11 dominant subspaces carry 46% of the total estimated intelligibility, while 115 dimensions carry 90%. Fig. 4 shows the basis vectors of the 11 dominant subspaces. The subspaces are characterized by regular spectro-temporal patterns, apparently with low spectro-temporal modulation frequencies.

Frequency analysis across the temporal dimension of each of the subfigures reveal temporal modulation frequencies—averaged across the acoustic frequency axis—ranging from 2.1 Hz to 5.8 Hz. This modulation frequency range is well-known to be particularly important for intelligibility. Specifically, Drullman *et al.* showed that intelligibility can be degraded significantly, if modulations in the frequency range of approximately 3–8 Hz are not preserved [26], [27]. Similarly, Elliot and Theunissen found temporal modulation frequencies in the range 1–7 Hz to be most important for speech intelligibility [24], while Kates and Arehart found frequencies less than 12.5 Hz to carry most information about intelligibility [28].

Applying Fourier transforms to the columns of each subspace spectrogram in Fig. 4 and computing the average magnitude spectrum shows maximum spectral modulations in the range 0.2–0.7 cycles/kHz. As for the temporal modulation content, these numbers are quantitatively well in line with the results in [24] who report spectral modulation frequencies < 1 cycle/kHz to be most important for speech intelligibility.[3]

While the eigenvalues λ_l of the sample correlation matrix $\frac{1}{2NM}(\check{X}\check{S}^T + \check{S}\check{X}^T)$ tend to be positive, a small subset of the lowest eigenvalues can be negative. In other words, the signal components represented by the corresponding subspaces *degrade* intelligibility as estimated by the model. For many practical situations, however, the impact of these negative intelligibility subspaces is small. For example, in Fig. 4, where the global SNR is -10 dB, the 15 smallest eigenvalues are negative—their sum is approximately -0.001, which is 0.3% of the total estimated intelligibility index. Generally speaking, the number and the impact of these negative intelligibility subspaces increases with decreasing SNR. For simple additive and stationary noise maskers, e.g., a single constant-frequency masker tone which occupy the same time-frequency region for all time segments, it can be verified that the time-frequency pattern of this masker may be represented well using the negative subspaces as basis functions. For non-stationary noise sources, on the other hand, e.g., the modulated low-pass noise used in Fig. 4, the negative subspaces do, generally, not represent the spectro-temporal noise pattern in a *particular* segment. Rather, the negative subspaces represent the *average* spectro-temporal pattern of the noise within many time segments, across which the noise does

[3]Note that an accurate comparison is difficult: the results in [24] are based on spectro-temporal analyses of log-magnitude spectra computed in a uniform-frequency filter bank, whereas the proposed method operates on linear, but energy-normalized one-third octave band magnitude spectra.

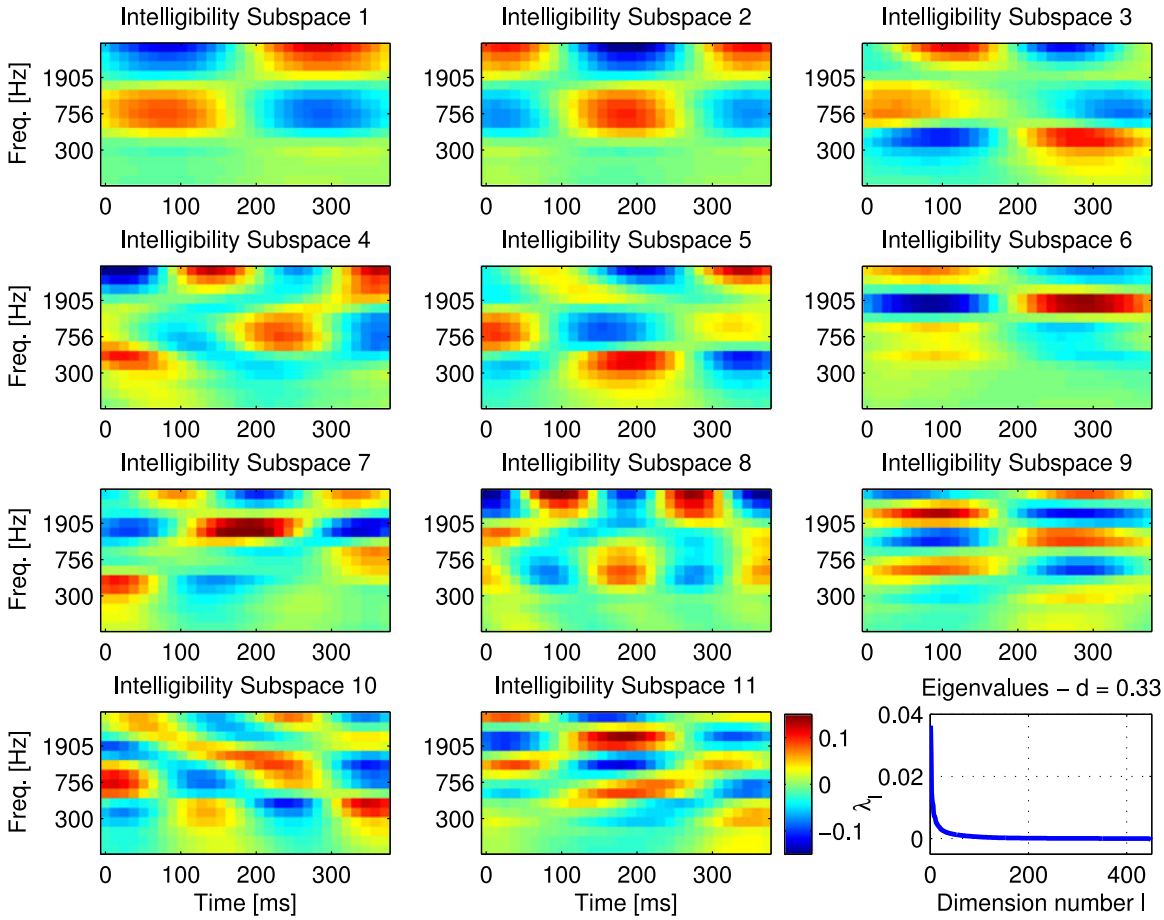


Fig. 4. Decomposition of d for speech in additive, speech-shaped, sinusoidally amplitude-modulated Gaussian noise ($f_{\mathrm{mod}} = 5$ Hz, SNR $= -10$ dB). Basis functions e_l of the 11 dominant intelligibility subspaces and decomposition of d in terms of eigenvalues λ_l (lower right).

not necessarily occupy the same time-frequency region in each time segment.

Finally, Fig. 5 shows the intelligibility subspace decomposition for speech in natural noise, namely a noise recorded in a busy office cafeteria [29]. While details obviously differ from the decomposition in Fig. 4, the main features of the dominant subspaces are the same: temporal modulation frequencies are in the range 2.0–5.7 Hz, while spectral modulations are in the range 0.2–0.8 cycles/kHz.

IV. Simulation Results

In this section, we present a number of intelligibility listening tests for evaluating the proposed method. Furthermore, we study the performance as a function of the segment length N and the test signal duration. Finally, we compare the performance of *ESTOI* to a range of existing speech intelligibility predictors.

A. Signals and Processing Conditions

We study the performance of *ESTOI* using the results of five intelligibility tests with speech signals subjected to various noise sources and processing conditions. The first two tests used various additive noise sources with strong temporal modulations; we include these in the study to verify the ability of *ESTOI* to operate in this domain, and to verify the results reported in [21] that established intelligibility predictors work less well here. The third test used stationary and non-stationary additive noise sources, with less temporal modulations; quite some existing methods work well for this common class of noise sources, and it is important to establish that *ESTOI* does so too. The fourth and fifth intelligibility test used processed noisy speech signals for which *STOI* works exceptionally well, while many other methods fail. As before, it is important to establish the performance of *ESTOI* in this situation.

1) Additive Noise Set I: The first set of signals consist of ten mainly non-stationary noise sources with significant modulation content, cf., Table I. The *icra* signals are synthetic speech signals constructed by filtering Gaussian noise sequences through bandpass filters with time-varying gain to construct signals with speech-like spectro-temporal properties [30]. The *snam* signals are 100% sinusoidally, intensity-modulated speech-shaped noise signals. The signals are constructed by point-wise multiplication of the unmodulated speech-shaped noise signal *icra1* with the modulation sequence

$$a(n) = 1 + \sin(2\pi\omega n + \phi),$$

where ω denotes the angular modulation frequency, and $\phi \in [-\pi; \pi[$ is a random phase-value, drawn independently for each

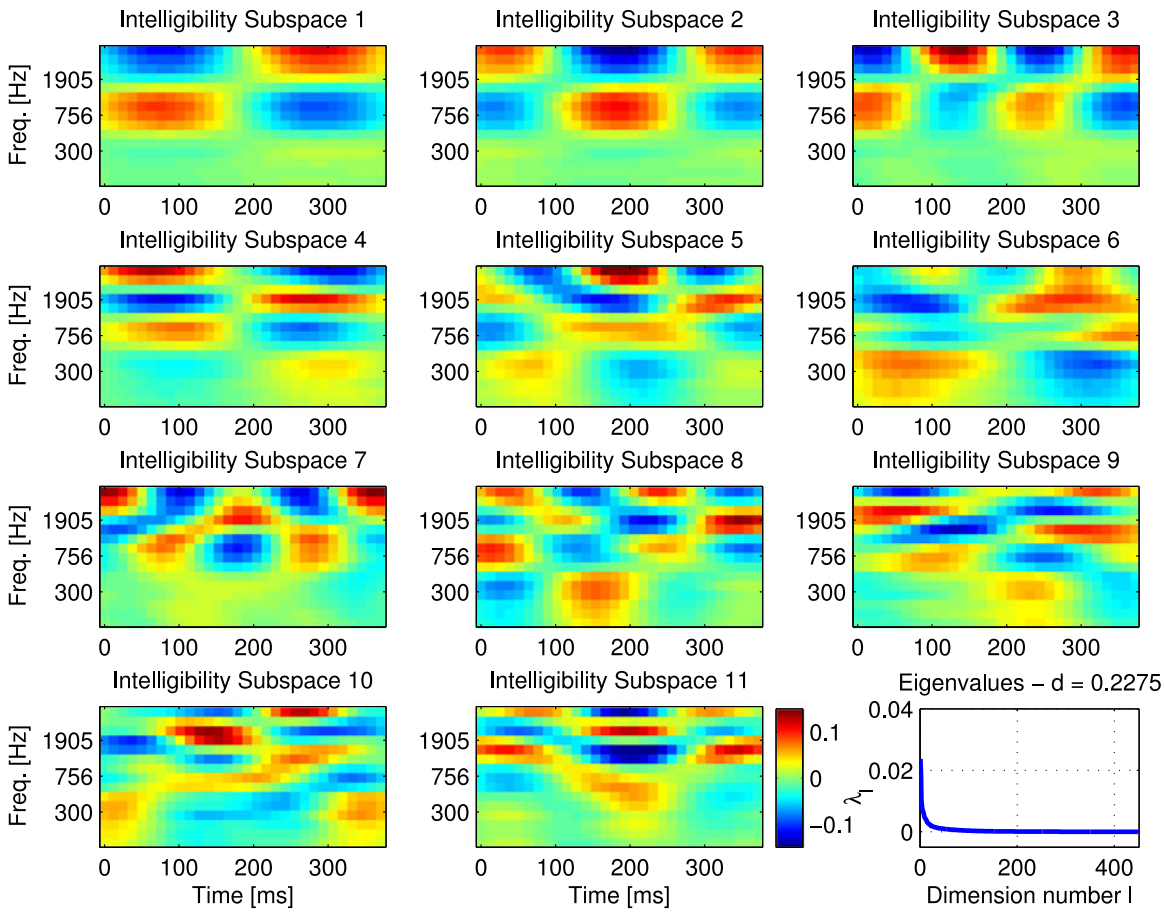


Fig. 5. Decomposition of d for speech in cafeteria noise [29], SNR $= -10$ dB. Basis functions e_l of the 11 dominant intelligibility subspaces and decomposition of d in terms of eigenvalues λ_l (lower right).

TABLE I
NOISE SOURCES AND SNR RANGES USED FOR INTELLIGIBILITY TEST WITH *Additive Noise Set I*

Noise Name	Description	SNR [dB]
icra1	Unmodulated speech-shaped (male) Gaussian noise from the ICRA corpus (Track 1) [30].	−17:3:−2.
icra4	1-person babble (female) from the ICRA corpus (Track 4).	−29:3:−14.
icra6	2-persons babble (1 male and 1 female) from the ICRA corpus (Track 6)	−24:3:−9.
icra7	6-persons babble from the ICRA corpus (Track 7)	−19:3:−4.
snam2	100% intensity-modulated versions of *icra1*. Modulation frequency 2 Hz.	−27:3:−12.
snam4	As above with modulation frequency 4 Hz.	−23:3:−8.
snam8	As above with modulation frequency 8 Hz.	−25:3:−10.
snam16	As above with modulation frequency 16 Hz.	−22:3:−7.
macgun	(Modified) machine gun noise from the Noisex corpus [31].	−37:3:−22.
destop	Destroyers operation room noise from the Noisex corpus [31].	−14:3:1.

Notation $x : y : z$ indicates SNRs from x to z (both included) in steps of y dB.

signal generation. To construct machine gun noise *macgun* with sufficient masking power, the original machine gun noise signal from the Noisex database [31] was divided into successive 20 ms frames, and frames with energy less than 40 dB of the maximum frame energy were removed.

Speech signals from the Dantale II sentence test [32] were added to randomly selected sections of each of the ten noise sources at six different SNRs (cf., Table I). The SNRs were chosen so that, for each noise source, some noisy signals were almost perfectly intelligible, whereas others were essentially unintelligible. The total number of conditions was therefore 10 noise types x 6 SNRs $= 60$ conditions. Each condition was repeated 3 times (with different speech and noise realizations) leading to a total of 180 sentences to be judged per subject. The presentation order of noise types, SNRs, and repetitions was randomized. The sample rate was 20 kHz.

We conducted a closed Danish speech-in-noise intelligibility test, cf., [33]. The Dantale II sentences consist of five words with a correct grammatical structure. Candidate words were arranged in an 10-by-5 matrix on a computer screen, such that each of the

five columns encompassed exactly the 10 possible alternatives for the corresponding word. Each column was extended with one entry, which allowed the subject to answer "Don't know". For each 5-word sentence, the subject must select via a graphical user interface the words that she heard. Subjects were seated in a sound treated room, where signals were presented diotically through headphones (Sennheiser HD 280 Pro). The *icra1* noise at the SNR of -8 dB was used to calibrate the presentation level to 65 dB (A). The subjects were allowed to adjust this level during a training session prior to the actual test. Twelve native Danish speaking subjects (normal-hearing, age range 26–44 years, 2 females, 10 males) participated in the test. The subjects volunteered for the experiments and were not paid for their participation.

2) Additive Noise Set II: The second data set, consisting of speech in additive fluctuating noise sources, is described in [7]. We use Maskers 1–13 [7], which include low-pass filtered unmodulated Gaussian noise, and various amplitude modulated Gaussian noise signals, including sinusoidally amplitude-modulated signals (modulation frequencies: 2.1, 4.9, 10.2, 19.9 Hz, and modulation depths of ± 6 dB, ± 12 dB, and $\pm 100\%$), and three irregularly modulated noise signals found by adding the sinus-modulators with random initial phases. The speech material used was the Swedish version of the Hagerman material [34], which is similar in structure to the Dantale 2 set used above. For each noise source, noisy speech signals were generated with an SNR of -15 dB, and the corresponding speech intelligibility was recorded ([7, Fig. 5]). Hence, the number of conditions equalled 11 noise sources $\times$ 1 SNR $=$ 11 conditions. Intelligibility tests were conducted with i) eleven young (17–33 years), normal hearing listeners, and ii) twenty elderly (54–69 years), normal hearing listeners (the study also included elderly, hearing-impaired listeners, but results of these tests are not used in this paper). The sample rate used was 20 kHz. For more details, we refer to [7].

3) Additive Noise Set III: We include a third additive noise set, for which many existing intelligibility predictors work well, see e.g. [14], [17] and the references therein. The data set encompasses Dantale 2 speech sentences contaminated by four additive noise sources: i) speech-shaped Gaussian noise, ii) car cabin noise recorded when driving on the highway, iii) bottling hall noise, and iv) cafeteria noise consisting of a conversation between a female and a male speaker, i.e., two-talker speech babble [35]. The noisy signals were generated with SNRs from -20 dB to 5 dB in steps of 2.5 dB, so the total number of conditions equals 4 noise types $\times$ 11 SNRs $=$ 44 conditions. Fifteen listeners participated in the test. For more details, we refer to [35].

4) Ideal Time-Frequency Segregation (ITFS): The fourth data set consists of the noisy signals from *Additive Noise Set III*, processed using the ITFS technique [36]. Kjems *et al.,* [35] processed noisy signals with two different ITFS algorithms called *ideal binary mask (IBM)* and *target binary mask (TBM)*, and used eight different variants of each algorithm (reflected by the *LC* parameter, i.e., the threshold for which the algorithm suppresses a given time-frequency tile or not). Three different SNRs were used, leading to a total number of (4 noise types (IBM) $+3$ noise types (TBM)[4]) $\times$ 8 LC $\times$ 3 SNRs $=$ 168 test conditions. Fifteen normal hearing subjects participated in the test. The sample rate was 20 kHz. More details are available in [35].

5) Single-Channel Noise Reduction: The last data set consists of noisy speech signals processed with three single-microphone noise reduction algorithms [37]. The three algorithms are all non-linear and aim at finding binary or soft minimum mean-square error estimates of the short-time spectral amplitude. We include this data set, because an obvious use of the proposed algorithm is for development/tuning of noise reduction algorithms. Speech-shaped (unmodulated) noise signals were added to speech signals (female speaker) from the Dutch version of the Hagerman test [33], [34] at fixed SNRs of -8, -6, -4, -2, and 0 dB. The noisy and processed speech signals were presented diotically via headphones, and the order of presenting the different algorithms and SNRs was randomized. The signals were evaluated in a closed Dutch speech-in-noise intelligibility test [33]. Each processing condition was repeated five times, leading to 4 conditions $\times$ 5 SNRs $=$ 20 test conditions. Thirteen subjects participated in the test. The sample rate was 8 kHz.

B. Prediction of Absolute Intelligibility and Figures of Merit

Most intelligibility prediction methods (including *ESTOI*) do not predict intelligibility, i.e., the fraction of words understood, *per se*. Instead, they output a scalar $\tilde{I}$ which, ideally, is monotonically related to absolute intelligibility I.[5] The monotonic mapping between predictor output and absolute intelligibility is generally hard to derive analytically, but in [14], [38] it was proposed to use the following logistic map,

$$\hat{I} = \frac{100}{1 + \exp\left(a\tilde{I} + b\right)}, \tag{10}$$

where $a, b \in \Re$ are constants that depend on the test material, test paradigm, etc., and which are estimated to fit the intelligibility data at hand.

To quantify the performance of intelligibility predictors, we use four figures of merit (see [17] for exact definitions): i) the linear correlation coefficient ρ_{pre} between average intelligibility scores obtained in listening tests, and the outcomes $\tilde{I}$ of the intelligibility predictors *before* applying the logistic map (see Eq. (10)), ii) the linear correlation coefficient ρ between average intelligibility scores and the outcomes $\hat{I}$ of the intelligibility predictors, i.e., after the logistic map, iii) the root mean-square prediction error σ between measured and predicted intelligibility, and iv) Kendalls rank correlation coefficient (τ).

C. Impact of Segment Length and Signal Duration

1) Sensitivity to Segment Length N: ESTOI was developed with simplicity in mind and has few free parameters. This section studies the performance of *ESTOI* as a function of the segment length N for the various noise/processing situations in the five

[4]The IBM and TBM algorithms are identical for speech-shaped noise.

[5]We use the symbol $\tilde{I}$ to represent the output of any intelligibility predictor (including *ESTOI*), while we reserve the symbol d for the particular scalar output produced by *ESTOI*.

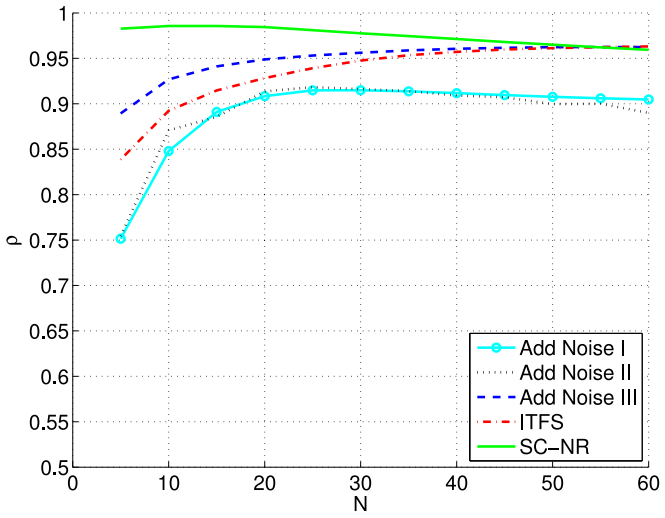


Fig. 6. Speech intelligibility prediction performance (in terms of ρ) as a function of segment length N for various noise/processing conditions. For *Additive Noise Set II*, we used the signals and listening test results for the young, normal-hearing listeners to avoid a cluttered plot. $N = 30$ corresponds to a segment duration of 384 ms.

data sets described above. In particular, for a given data set, and for a given choice of the segment length N, the proposed method was applied to compute an intelligibility index for each test condition in that data set. Then, the free parameters a, b, of the logistic function, Eq. (10), were fitted to map the predicted intelligibility indices to absolute intelligibility as measured in a listening test. Finally, the performance in terms of ρ, σ, and τ was computed.

Fig. 6 shows the performance in terms of ρ as a function of segment length N (in the range $N = 5, \ldots, 60$ corresponding to durations of 64–768 ms). Clearly, the proposed method is fairly insensitive to the exact choice of the segment length N. In fact, for $20 \leq N \leq 50$ (corresponding to durations of 256–640 ms), the proposed method gives excellent performance with values of $\rho > 0.9$. The lower performance for $N < 20$ may be explained by the fact that with these short segment lengths, the method is less able to capture low-frequency temporal modulations, which are important for speech intelligibility [14]. While the discussion above focused on prediction performance in terms of ρ, similar conclusions may be drawn from performance analysis based on σ and τ (not shown). Based on these observations, a value of $N = 30$ (384 ms) is used in the remainder of the simulation experiments.

2) Sensitivity to Duration of Test Signals: For highly modulated, additive noise sources, the instantaneous SNR can vary significantly across a short time span. For example, a sinusoidally amplitude-modulated noise source could completely mask the target signal at one instant, while leaving it essentially un-masked half a period later (cf., Fig. 3). Hence, the speech intelligibility for a particular short speech signal is highly dependent on the (random) location of the high SNR region with respect to the speech signal. Since our goal is to estimate the *average* speech intelligibility, we would therefore expect it to be necessary to average across *many* noise realizations, or, equivalently, to use longer test speech signals, than would e.g. be

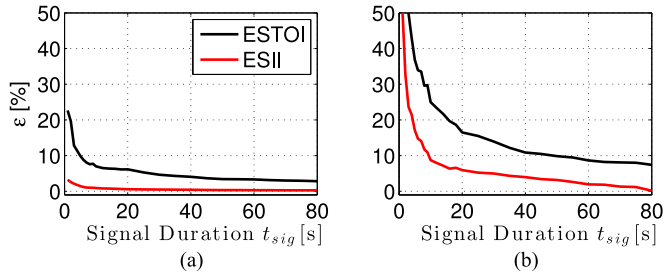


Fig. 7. Relative standard deviation $\epsilon(t_{\text{sig}})$ of speech intelligibility predictors *ESII* and *ESTOI*, as a function of test signal duration t_{sig}. (a) Speech-shaped stationary noise (SNR $= -10$ dB), (b) *icra4*-noise (synthetic 1-person babble) [30] (SNR $= -23$ dB).

necessary for unmodulated noise sources. In this section we therefore study the sensitivity of the proposed method with respect to test signal duration t_{sig}.

To do so, we generated speech signals contaminated by two additive noise sources: speech-shaped stationary noise, and synthetic 1-person babble (the *icra4* noise source from [30]). The noise sources were scaled to achieve SNRs of -10 dB and -23 dB, respectively, corresponding approximately to the 50% speech reception threshold (SRT) (the SNR needed to achieve a recognition rate of 50%) for these noise sources. Noise and clean signals were generated in corresponding pairs with various durations in the range 1–80 secs. Noisy signals were generated by adding the clean and noise signals. Clean and noisy signals were then passed through *ESTOI*. For comparison, the clean and noise signals were passed through the *ESII* algorithm [3] (see Section IV-D for implementational details). For each signal duration, $n_{\text{real}} = 100$ different realizations of the clean/noisy signal pairs were evaluated.

It is of interest to study to which extent an intelligibility prediction $\tilde{I}_n(t_{\text{sig}})$ based on a single (the nth) test signal realization of duration t_{sig} lies in the neighborhood of the ensemble-average, i.e., the average predictor value $\mu_{\tilde{I}(t_{\text{sig}})} = \frac{1}{n_{\text{real}}} \sum_n \tilde{I}_n(t_{\text{sig}})$ across many realizations n_{real} of test signal pairs. To do so, let us define the sample standard deviation

$$\sigma_{\tilde{I}(t_{\text{sig}})} = \sqrt{\frac{1}{n_{\text{real}}} \sum_n \left(\tilde{I}_n(t_{\text{sig}}) - \mu_{\tilde{I}(t_{\text{sig}})} \right)^2},$$

and let us define the relative standard deviation as

$$\epsilon(t_{\text{sig}}) = \sigma_{\tilde{I}(t_{\text{sig}})} / \mu_{\tilde{I}(t_{\text{sig}})} \times 100 \quad [\%].$$

Fig. 7(a) shows $\epsilon(t_{\text{sig}})$ for *ESII* and *ESTOI* for unmodulated speech-shaped noise, while Fig. 7(b) shows the results for babble noise. From Fig. 7 three conclusions can be drawn. First, as expected, the relative standard deviation declines with signal duration. Secondly, for a given test signal duration, *ESII* has lower relative standard deviation than *ESTOI*. This is because *ESII* makes explicit use of its access to the clean speech signal and the noise signal in separation to accurately compute SNRs in different time-frequency regions, and subsequently compute an intelligibility index based on these (i.e,. *ESII* is a Class-1 method as discussed in the Introduction). *ESTOI*, on the other hand, does not make use of the access to the separated clean

TABLE II
INTELLIGIBILITY PREDICTORS FOR COMPARISON

Acronym	Short Description
ESTOI	Extended Short-Time Objective Intelligibility (*ESTOI*)—Proposed method.
SIMI	Speech Intelligibility prediction based on Mutual Information [17].
STOI	The short-time objective intelligibility measure [14].
CSII-MID	The mid-level coherence speech intelligibility index (SII) [9].
CSII-BIF	The coherence SII with signal-dependent band importance functions [39]).
STI-NCM	The normalized covariance speech transmission index (STI) [10].
STI-NCM-BIF	The normalized covariance STI with signal-dependent band-importance functions [39]).
NSEC	The normalized subband envelope correlation method [45].
MIKNN	Intelligibility prediction based on a k-nearest neighbor estimate of mutual information (MIKNN) [46].
GLIMPSE*	Implementation of Cooke's glimpse method [8].
SII*	The Critical-Band SII with SPIN band-importance functions ([3, Table B.1]).
ESII*	Implementation of Extended SII [6] ([3, Table B.1]).

Note that predictors marked with $*$ require that speech and noise realizations are available in separation, and that noise is additive, i.e., these methods cannot be directly used to predict the intelligibility of (non-linearly) processed speech signals.

TABLE III
PERFORMANCE OF INTELLIGIBILITY PREDICTORS IN TERMS OF ρ_{pre}, I.E., THE LINEAR CORRELATION COEFFICIENT BETWEEN MEASURED INTELLIGIBILITY AND PREDICTOR OUTPUTS $\tilde{I}$, CF., (10)

	Add. Noise Set I	Add. Noise Set II (Young)	Add. Noise Set II (Elderly)	Add. Noise Set III	ITFS	SC-NR
ESTOI	0.846	**0.877**	**0.900**	0.864	0.919	0.955
SIMI	0.483	−0.028	−0.253	**0.931**	**0.934**	0.974
STOI	0.477	−0.797	−0.789	0.887	0.931	**0.983**
CSII-MID	0.671	0.002	−0.020	0.784	0.440	0.794
CSII-BIF	0.717	0.578	0.696	0.883	0.541	0.843
STI-NCM	0.480	−0.354	−0.657	0.880	0.731	0.844
STI-NCM-BIF	0.519	−0.033	−0.153	0.784	0.568	0.974
NSEC	0.572	−0.309	−0.349	0.924	0.871	0.638
MIKNN	0.552	0.755	0.770	0.732	0.824	0.847
GLIMPSE	**0.850**	0.851	0.850	0.845	–	–
SII	0.541	−0.101	−0.112	0.723	–	–
ESII	0.807	0.816	0.849	0.701	–	–

TABLE IV
PERFORMANCE OF INTELLIGIBILITY PREDICTORS $\hat{I}$ IN TERMS OF LINEAR CORRELATION COEFFICIENT ρ

	Add. Noise Set I	Add. Noise Set II (Young)	Add. Noise Set II (Elderly)	Add. Noise Set III	ITFS	SC-NR
ESTOI	**0.915***	**0.895***	**0.916***	0.960*	0.948*	0.981*
SIMI	0.514	0.027	0.238	0.975*	0.958*	0.970*
STOI	0.477	0.809*	0.799*	0.967*	**0.961***	**0.987***
CSII-MID	0.766	0.000	0.018	0.948	0.454	0.799
CSII-BIF	0.745	0.607*	0.791*	**0.978***	0.539	0.850
STI-NCM	0.525	0.364	0.660*	0.935	0.727	0.844
STI-NCM-BIF	0.524	0.003	0.147	0.813	0.586	0.976*
NSEC	0.619	0.315	0.356	0.953*	0.868	0.625
MIKNN	0.722	0.780*	0.808*	0.902	0.881	0.870
GLIMPSE	0.872*	0.872*	0.875*	0.912	–	–
SII	0.674	0.098	0.107	0.964*	–	–
ESII	0.845	0.844*	0.872*	0.818	–	–

Superscripts * indicate methods which do not perform statistically significantly worse than the method with the highest ρ for a given data set (see text for details).

and noise components and is therefore more generally applicable, e.g., to non-linearly processed signals (i.e., it is a Class-2 method). This generality comes with the price of an increase in the estimation standard deviation. Thirdly, as expected, for a given test signal duration, the estimation standard deviation is higher for modulated than for non-modulated noise, both for *ESII* and *ESTOI*.

It is hard to decide *a priori* on a sufficient test signal duration, because a) this depends on the noise signal statistics in a non-trivial manner, and b) the noise statistics are unavailable to the proposed method. Hence, the test signal duration should simply be chosen as long as practically possible, and generally no less than some tens of seconds. Note that long test signals can be generated by concatenating several of the, potentially short, speech sentences used in the intelligibility test.

D. Comparison to Existing Methods

We compare the proposed intelligibility prediction method to reference methods from the literature. The methods are outlined in Table II. The methods *CSII-BIF* and *STI-NCM-BIF* are referred to as $\mathrm{CSII}_{\mathrm{mid}}$, $W_4, p = 1$ and NCM, $W_i^{(1)}, p = 1.5$, respectively, in [39, Table IV]. We implemented the *GLIMPSE* method [8] using the one-third-octave filter bank used in *ESTOI*. The *speech glimpse percentage* was defined here as the percentage of time-frequency units with a local SNR exceeding -8 dB (this threshold was chosen because it lead to best performance in terms of ρ, σ, and τ). Our implementation of the *ESII* algorithm computes the per-frame SII based on one-third octave filtering, and outputs the average of the per-frame SIIs. The implementation uses stationary speech-shaped Gaussian noise instead of undistorted real speech signals as input (as specified in [3], [6]), but excludes the upward-spread-of-masking functionality as defined in [3], because this appears to degrade performance. As in [6], we use the band importance functions derived for the test stimuli of the Speech in the Presence of Noise (SPIN) test ([3, Table B.1]) and [40].

Tables III–VI summarize performance in terms of ρ_{pre}, ρ, σ, and τ, respectively, for the intelligibility predictors for the various additive noise and processing conditions. The ρ, σ, and τ, values are found by fitting a, b to the data sets in question. To identify statistically significant differences between ρ-values (see Table IV), pairwise comparisons using the Williams t-test [41]–[43] were performed within each data set between the predictor with the largest ρ and the others (Bonferroni correction for multiple comparisons). Methods which do not perform statistically significantly worse than the method with highest ρ ($p < 0.05$) are indicated with (*) in Table IV. In addition, a statistical analysis of significance was applied to the root mean-square prediction errors σ (see Table V) as follows (see [44] for a brief outline of this approach). For each prediction method and for each of the five listening tests, the free parameters a, b in the logistic function were fitted to $n - 1$ data points, where n denotes the number of conditions for a specific data set. Then this logistic function was applied to the

TABLE V
PERFORMANCE OF INTELLIGIBILITY PREDICTORS $\hat{I}$ IN TERMS OF ROOT MEAN-SQUARE PREDICTION ERROR σ

	Add. Noise Set I	Add. Noise Set II (Young)	Add. Noise Set II (Elderly)	Add. Noise Set III	ITFS	SC-NR
ESTOI	**11.49***	**7.63***	**7.03***	10.26*	9.93*	3.76*
SIMI	24.92	17.09	17.01	8.13*	8.91*	4.72*
STOI	25.53	10.06*	10.54*	9.24*	**8.61***	**3.09***
CSII-MID	18.69*	17.10	17.50	11.47*	27.72	11.65
CSII-BIF	19.39*	13.62*	11.03*	**7.54***	26.23	10.20
STI-NCM	24.78	15.93*	13.18*	12.75*	21.39	10.38
STI-NCM-BIF	24.74	17.09	17.32	21.08	25.23	4.22*
NSEC	22.89	16.23*	16.36*	10.89*	15.51	15.12
MIKNN	20.19*	10.74*	10.37*	15.92*	14.68	9.55
GLIMPSE	14.22*	8.37*	8.51*	14.85*	–	–
SII	21.56	17.02	17.40	9.76*	–	–
ESII	15.59*	9.20*	8.59*	20.82	–	–

Superscripts * indicate methods which do not perform statistically significantly worse than the method with the lowest σ for a given data set (see text for details).

TABLE VI
PERFORMANCE OF INTELLIGIBILITY PREDICTORS IN TERMS OF $\hat{I}$ KENDALL'S RANK CORRELATION COEFFICIENT τ

	Add. Noise Set I	Add. Noise Set II (Young)	Add. Noise Set II (Elderly)	Add. Noise Set III	ITFS	SC-NR
ESTOI	**0.748**	**0.590**	**0.667**	0.816	0.775	0.842
SIMI	0.354	0.077	0.256	0.825	0.767	0.884
STOI	0.376	0.436	0.436	0.818	**0.798**	**0.905**
CSII-MID	0.609	0.026	0.026	**0.863**	0.335	0.600
CSII-BIF	0.580	0.410	0.487	0.846	0.408	0.684
STI-NCM	0.328	0.180	0.590	0.818	0.538	0.684
STI-NCM-BIF	0.276	0.085	0.077	0.643	0.385	0.853
NSEC	0.514	0.077	0.128	0.854	0.695	0.505
MIKNN	0.554	0.436	0.410	0.751	0.689	0.684
GLIMPSE	0.673	0.461	0.385	0.742	–	–
SII	0.260	0.330	0.128	0.822	–	–
ESII	0.653	0.564	0.359	0.685	–	–

left-out data point $\tilde{I}_i$, where i is the index of the left-out data point, to find a prediction $\hat{I}_i$ of the left-out subject result I_i. The procedure was repeated for all data points, resulting in prediction errors $e_i = I_i - \hat{I}_i, i = 1, \ldots, n$. Our goal is to compare the magnitude of these prediction errors across prediction methods. The data e_i^2 for each intelligibility predictor and for each data set did not pass a chi-square goodness-of-fit test for normality ($p < 0.05$). Hence, a Kruskal–Wallis test was performed, rejecting for each data set the hypothesis that the median of e_i^2 is identical for all prediction methods ($p < 10^{-5}$). A multiple pairwise comparison test (Tukey HSD) was applied to identify prediction methods which, for a particular data set, performed statistically significantly worse than the method with lowest σ ($p < 0.05$). The result of this comparison is indicated with (*) in Table V.

From these tables, a number of observations can be made. First, focusing on the highly non-stationary noise conditions, i.e., *Additive Data Sets I* and *II*, it is clear that *ESTOI*, *GLIMPSE*, and *ESII* appear to work quite well. The fact that *GLIMPSE* and *ESII* can work well in these conditions is well in line with results reported in [8] and [6], respectively. On the other hand, existing methods such as *STOI* and *SIMI*, which are known to work well for other less non-stationary noise sources and for various processing conditions, do not work well for the highly fluctuating noise sources: *SIMI* shows correlation values $\rho \approx 0$, while *STOI* shows large but negative correlations for these data sets in Table III. This indicates that the STOI output $\tilde{I}$ *decreases* for increasing measured intelligibility (the fact that the same entry in Table IV is positive is because the logistic map from $\tilde{I}$ to $\hat{I}$ in this situation maps low $\hat{I}$ values to high $\tilde{I}$ values, and vice versa).

It is interesting to note that *ESTOI* (and *ESII* and *GLIMPSE*), perform well for *Additive Noise II*, both for young and for elderly normal-hearing subjects. While the basic intelligibility predictors are unchanged, each intelligibility predictor employs a different logistic map (i.e., constants a and b in Eq. (10)) for the different subject groups, because the 30% SRT was 6 dB higher for the elderly compared to the young subjects. It appears that the SRT differences between these normal-hearing subject groups (e.g., differences in higher auditory stages, which are not captured by a standard listening test used to establish whether a subject is normal hearing or not) are well-modeled simply by changing the logistic map.

Secondly, for the less fluctuating (but still non-stationary) noise sources in *Additive Noise Set III*, most methods work well. In fact, for this data set, several intelligibility predictors, including *ESTOI*, show values of $\rho > 0.95$, and σ values at or below 10%. Note that *SII*, which relies on long-term noise spectra, also works well in this situation.

For noisy signals processed with ITFS and single-channel noise reduction, *ESTOI*, *SIMI*, and *STOI* work well with $\rho > 0.94$. It is interesting to note that for single-channel noise reduced signals, *STI-NCM-BIF* works exceptionally well ($\rho > 0.97$): an explanation is that *STI-NCM-BIF* was developed with this particular processing type in mind; also note that *STI-NCM-BIF* does not show this level of performance for any other noise/processing condition.

In summary, for highly fluctuating, additive noise sources, where *STOI* and *SIMI* fail, *ESTOI* performs at the level of established methods such as *GLIMPSE* and *ESII*, without requiring access to the speech and noise signals in isolation. For less fluctuating noise sources, *ESTOI* performs as well as the best existing methods, such as *SII*, *STOI*, and *SIMI*. For non-linearly processed noisy signals, where methods such as *SII*, *ESII*, and *GLIMPSE* cannot operate, the proposed method still performs at the level of *STOI* and *SIMI*.

V. CONCLUSION

We presented an algorithm for monaural, intrusive intelligibility prediction: given undistorted reference speech signals and their noisy, and potentially non-linearly processed, counterparts, the algorithm estimates the average intelligibility of the latter, across a group of normal-hearing listeners. The proposed algorithm, which is called *ESTOI*, may be interpreted in terms of an orthogonal decomposition of energy-normalized short-time spectrograms into "intelligibility subspaces", i.e., one-dimensional subspaces which are ranked according to

their importance wrt. intelligibility. This intelligibility subspace decomposition indicates, that the proposed algorithm favors spectro-temporal modulation patterns, which are known from literature to be important for intelligibility. The proposed intelligibility predictor has only one free parameter, the segment length N, i.e., the duration across which the short-time spectrograms are computed. We show, via simulation experiments, that performance is fairly insensitive to the exact choice of this parameter, and that durations in the range 256–640 ms lead to best performance—this allows the algorithm to capture relatively low-frequency modulation content, while still being able to adapt to changing signal characteristics. We study the performance of *ESTOI* in predicting the results of five different intelligibility listening tests: two with temporally highly modulated additive noise sources, one with more moderately modulated, additive noise sources, and two with noisy signals processed by ideal time-frequency masking and single-channel non-linear noise reduction algorithms, respectively. Compared to a range of existing speech intelligibility prediction algorithms, *ESTOI* performs well across all listening tests.

The present study has focused on speech intelligibility prediction performance within data sets, that each contain signals with similar distortions or processing types (e.g., additive, modulated noise or noisy speech processed by ITFS algorithms, etc.). It is a topic for future research to study the performance of the proposed intelligibility predictor *across* data sets with different distortion and processing types. Compared to the present study, this would require conduction of larger intelligibility tests, where these different distortion or processing types are included in the same listening test.

A MATLAB implementation of the proposed algorithm is available for non-commercial use at http://kom.aau.dk/∼jje/.

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